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Impact of Pellet Thickness on Quantitative Terahertz Spectroscopy of Solid Samples in a Polyethylene Matrix

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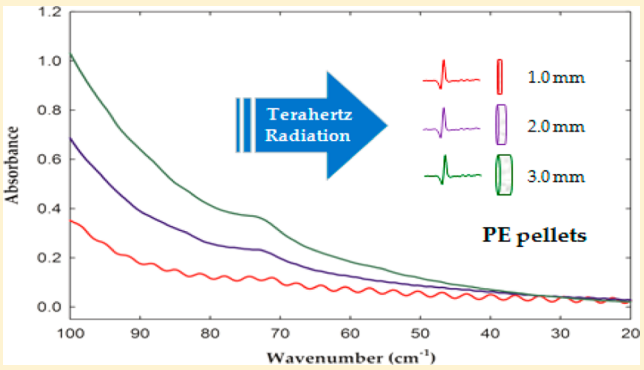
Supporting Information

ABSTRACT: Pellets composed of different weight-percent (wt-%) of lactose within a polyethylene (PE) matrix are used to examine how the physical thickness of solid samples impact analytical measurements performed over terahertz (THz) frequencies when using time-domain THz spectroscopy. Results indicate that the thickness of each pellet depends on the mass and physical properties of the individual components that comprise the pellet. Thickness of mixture pellets depends on the porosity of the individual pellet components. Porosity measurements presented here for PE and lactose give values of 25.6 ± 0.3 and 14.5 ± 0.1, respectively, which indicate that more air is trapped within the compressed PE matrix compared to that for lactose. This difference in porosity creates different pellet thicknesses for pellets of the same nominal mass but with different relative amounts of PE and lactose. For this binary matrix, the thickness of each pellet is found to be a linear combination of the compressed densities of the individual components. Analysis of the time-domain THz spectra reveals that thinner samples are confounded by a fringe pattern observed in the frequency-domain spectra. This fringe pattern is created by an etalon corresponding to the air/pellet interfaces for the sample in the optical path. Spectra collected from thicker pellets are confounded by a sloping baseline caused by scattering effects within the pellet matrix. The quantitative impact of pellet thickness is determined by comparing the mean standard error of calibration (MSEC) and mean standard error of prediction (MSEP) for a set of leave-three-out cross validation multivariate calibration models based on the partial least-squares (PLS) algorithm. Results indicate that PLS models are capable of analytical measurements with MSEC and MSEP values between 0.04 and 0.20 wt-%. Analysis of spectral variance captured within the corresponding spectral loadings for each model indicates that spectral variance is lowest for the 300 mg samples where the impact of scattering is minimal under conditions when the sample etalon is nonexistent.



Terahertz (THz) spectroscopy has been proposed as an analytical tool for the quantification of components in solid samples.[1−3] This recent interest in quantitative THz spectroscopy has been driven by advances in semiconductor materials that enable efficient generation and detection of electromagnetic (EM) radiation over THz frequencies.[4] These materials are central to successful commercial THz spectrometers designed for collecting spectra over the 150−10 cm^{-1} range where transitions correspond to rotational modes in gases and intermolecular interactions in solids.

THz time-domain spectroscopy (THz-TDS) is a common method for collecting THz spectra with high signal-to-noise ratios (SNR's). As detailed elsewhere, broadband THz radiation is produced when a femto-second pulse of 800 nm radiation strikes a biased GaAs or ZnTe semiconductor antenna.[1,5−7] The produced THz radiation is coherent and passes through the sample, after which the magnitude of the electric-field vector of the transmitted EM radiation is detected as a function of time. A Fourier transformation is used to convert the resulting time-domain spectrum to a frequency-domain spectrum.

Solid samples are commonly prepared for THz measurements by mixing the sample with polyethylene (PE) powder and pressing this mixture into a pellet under pressure. PE is well suited for such pellets because of its low absorptivity over THz frequencies and the ability to form stable pellets under moderate pressures.[8] Many reports use PE as the pellet matrix for analytical measurements of different types,[9−11] including the quantification of active ingredients in drug formulations[12,13] and identification of polymorphs in pharmaceutical products.[14−18]

Transmission THz measurements through PE-based samples are sensitive to an array of experimental parameters, including orientation of the molecules relative to the direction of the EM wave,[8] particle size,[19−21] porosity,[22−25] and heterogeneity[18,19,26]

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of the pressed pellet. Despite efforts to thoroughly mix the solid sample material within the PE powder prior to pellet formation, spectral artifacts are created owing to the physical nature of the pellet. These spectral artifacts can be superimposed on the spectral features of the analyte, thereby confounding interpretation. The inability to control these physical parameters results in poor analytical performance as demonstrated by scatter within reported absorbance versus concentration plots.

Our initial efforts to explore the analytical utility of THz-TDS have focused on establishing methods for collecting reproducible spectra, thereby enhancing the potential for effective quantitative measurements. Thickness of the PE pellet is a critical parameter for analytical measurements based on transmission spectra.[27] Thicker pellets of the same concentration will enhance the analytical sensitivity by providing more molecules in the optical path. Some limit to pellet thickness must be realized, however, on the basis of either practical or spectroscopic considerations. The following results illustrate the influence of pellet thickness by characterizing spectra collected from a series of pellets composed of different weight-percent (wt-%) of lactate within a PE binder. An optimum thickness is identified for the quantification of lactose in a PE pellet matrix by partial least-squares (PLS) analysis.

■ EXPERIMENTAL SECTION

Pellet Preparation. Pellets were prepared by placing the desired mass of the powder sample into a 13 mm diameter stainless-steel die. Prior to applying pressure, a vacuum was applied for three minutes to remove trapped air. While maintaining the vacuum, a constant pressure of 5 tons was applied to the powder for two minutes. After releasing the pressure and removing the vacuum, the pellets were carefully removed from the die. Upon removal, the pellet disks were allowed to relax for a period of 30 min, after which no further changes in the dimensions of the disk were evident.

Blank PE pellets were prepared from dry PE powder purchased from Sigma-Aldrich, Milwaukee, WI. The PE powder had a listed particle size range of 53−75 μm and was stored in a desiccator until use. Blank PE pellets were prepared with different masses of PE and different pressures, as detailed below.

Pellets with lactose were prepared by pressing thoroughly mixed samples of PE and lactose powder. The lactose was purchased from Sigma-Aldrich as β-lactose powder and a sieve was used to isolate powders composed of particles with diameters less than 50 μm. Pellets were prepared with 11 different concentrations of lactose ranging from 5.0 to 15.0 wt-% by increments of 1.0 wt-%. For each lactose concentration, a set of seven different pellets was prepared with different thicknesses by varying the total mass of pellet material from 100 to 400 mg in increments of 50 mg.

Acquisition of THz Spectra and Data Processing. Data were collected as time-domain THz spectra by passing the incident radiation through the center of the pellet. Each time-domain spectrum was collected with a TPI Spectra1000 spectrometer (TeraView, Cambridge, UK) with a resolution setting of 1.2 cm^{-1}. A detailed description of the instrumentation can be found elsewhere.[1,17,28] Each spectrum was obtained by coadding 1800 scans over approximately one minute. The sample chamber was purged with dried air throughout to avoid absorption features associated with gaseous water. The collected time-domain signal was initially treated with a Blackman-Harris 4-term apodization function and then Fourier transformed to yield the corresponding single-beam spectrum. Absorbance spectra were calculated as the negative logarithm of the ratio of the single-beam spectrum of the sample divided by the single-beam spectrum of either air or a blank PE pellet, as indicated.

In order to reduce the impact of pellet heterogeneity, six spectra were collected for each pellet. The first three spectra were collected through one face of the pellet by manually rotating the pellet by 120° between each spectrum. The pellet was then flipped to expose the opposite face of the pellet to the incident radiation and three additional spectra were collected with the same 120° rotations. The average of these six spectra was taken as a single sample spectrum for each pellet.

Lactose levels were quantified by using PLS regression. PLS calibration models were generated from the resulting absorbance THz spectra. The PLS algorithm was implemented in Matlab Version 7.0 (MathWorks Inc., MA).

■ RESULTS AND DISCUSSION

Pure Pellets. Initially, an array of 49 pellets containing only PE was prepared by pressing pellets with seven different amounts of PE powder at seven different pressures. Table S1 summarizes the different pellet fabrication conditions as well as the measured thickness for each pellet (Supporting Information). As indicated in Table S1, the magnitude of the applied pressure does not affect the thickness of the relaxed pellets. No trend in pellet thickness is evident across applied pressures and the standard deviation for each pellet thickness ranges from only 10 to 20 μm with relative standard deviations less than 2% across each mass. All subsequent pellets were prepared by using a pressure of 5 tons.

Pellet thickness is linearly dependent on the mass of powder. A regression analysis of the average values presented in Supporting Information Table S1 (thickness vs mass) reveals a slope of 10.49 ± 0.03 μm/mg and a coefficient of determination (R^2) of 0.99996 when the y-intercept was set to zero. Similarly, pellets containing only lactose were prepared with a 5 ton pressure and the thickness of the pellets measured for different amounts of lactose. A plot of thickness versus mass for the lactose data revealed a linear relationship with a slope of 5.53 ± 0.01 μm/mg and a coefficient of determination (R^2) of 0.99996 when the y-intercept was set to zero.

Thickness of these pellets depends on the density of the material and the amount of air trapped within the compressed material. The percent porosity gives a measure of the trapped air within the pellet matrix and characterizes the volume of a pellet as a function of mass. The percent porosity, $\varepsilon(\%)$, is calculated by eq 1 where ρ_{pellet} corresponds to the density of the pellet formed from the compressed material and ρ_{true} corresponds to the true, inherent density of the pure material. True densities were taken as 0.94 and 1.59 g/cm^3 for PE and lactose, respectively.[29,30]

$$\varepsilon(\%) = \left(1 - \frac{\rho_{pellet}}{\rho_{true}}\right) \times 100 \quad (1)$$

Densities of the PE and lactose pellets were determined from the dimensions and mass of the pure PE and lactose pellets, as indicated in the Supporting Information. The measured compressed densities of PE and lactose are 0.698 ± 0.002 and 1.366 ± 0.004 g/cm^3, respectively. According the eq 1, the percent porosities of the pressed PE and lactose pellets are 25.6 ± 0.3 and 14.5 ± 0.1, respectively, for the pressed PE and

Table 1. Thickness[a] of Pellets Containing Different Relative Amounts of PE and Lactose

	total pellet mass						
	100 mg	150 mg	200 mg	250 mg	300 mg	350 mg	400 mg
No lactose	1.07	1.56	2.07	2.65	3.15	3.67	4.19
5 wt-% lactose	1.03	1.53	2.00	2.53	3.12	3.55	4.06
10 wt-% lactose	1.02	1.50	2.00	2.47	3.02	3.42	3.96
15 wt-% lactose	0.99	1.45	1.93	2.42	2.96	3.38	3.90

[a]All thickness values reported in mm.

lactose pellets. These values indicate that the PE pellets contain more trapped air within the compressed matrix. Linearity of the density plots in Figure S1 (Supporting Information) indicates the percent porosity is independent of the mass of material used to prepare the pellet. This independence of porosity on mass is consistent with previous reports detailing the physical properties of compressed pellets designed as tablets for pharmaceutical applications.[22−24] While previous studies indicate that percent porosity of such pellets is dependent on the size of particles,[25] particle size is not a parameter examined in the present investigation. Still, it is noteworthy that different sources of PE can produce slightly different spectral features, most likely from different particle sizes and porosities (amount of trapped air).

Mixture Pellets. Pellets were prepared from mixtures of PE and lactose by using 5 tons of pressure and dimensions of the pellets were measured following relaxation upon removal from the die. Table 1 summarizes pellet thickness values determined for a set of PE pellets containing zero, 5, 10, and 15 wt-% lactose. As illustrated by the values in Table 1, pellet thickness decreases with a higher percentage of lactose. This result is consistent with the higher density and lower porosity of compressed lactose compared to compressed PE. As a fraction of the PE is replaced by lactose, the overall porosity of the pellet decreases, which results in a lower volume and thinner pellet.

As will be shown below, quality of a THz spectrum can be impacted by the physical thickness of the sample used in a transmission experiment. For this reason, it is important to understand the parameters that impact pellet thickness. As noted above, pellet thickness depends on the porosity of the materials embedded within the pellet composition. Equation 2 gives the functional relationship between pellet thickness and mass of each component under the assumption that the total thickness is the sum of the thickness of the individual components (see Supporting Information for details). Equation 2 is provided for pellets composed of mixtures of lactose and PE, where t_{pellet} is the thickness of the compressed pellet. In this expression, m_{lact}, and ρ_{lact}, correspond to the mass and compressed density of lactose, respectively, m_{PE}, and ρ_{PE}, correspond to the mass and compressed density of PE, respectively, and r corresponds to the radius of the compressed disk.

$$t_{pellet} = \left[\frac{m_{lact}}{\rho_{lact}\pi r^2}\right] + \left[\frac{m_{PE}}{\rho_{PE}\pi r^2}\right] \quad (2)$$

The validity of eq 2 was tested by comparing the measured and calculated thickness for a series of pellets prepared with different weight-percentages of lactose in a PE matrix. The results of this analysis are presented in Figure S2 (Supporting Information). The correlation plot in Figure S2 demonstrates the linear relationship between values calculated for eq 2 with slope, y-intercept, and coefficient of determination (R^2) values of 1.0066 ± 0.0036, 0.0003 ± 0.0010 mm, and 0.9991, respectively. The high degree of linearity of this plot combined with the closest of the regression coefficients to the ideal values for a unity correlation verifies the validity of eq 2. The standard error can be computed as the square root of the sum of squares of differences (calculated minus measured) divided by the number of samples. The standard error is 0.037 mm for the data presented in Figure S2.

Spectral Features of PE Pellets. The low absorptivity of PE over THz frequencies results in simple THz spectra and makes this polymer well suited as a matrix for solid samples. A series of time-domain and frequency-domain spectra is presented in Figure 1 for blank pellets of PE. For clarity and to assist comparisons, the time-domain spectra are offset vertically in Figure 1a, whereas, no offset is included in the frequency-domain spectra presented in Figure 1b.

The THz time-domain spectra in Figure 1a are similar for each PE pellet tested. As expected, these time-domain spectra feature a main peak corresponding to constructive interference from the broadband coherent THz source, followed by a decay pattern. A small feature is denoted at a time delay position of 15.3 mm for the 1.07 mm thick pellet. This same feature is present in the other time-domain spectra but shifts to longer delay times in accordance with thicker pellets. This feature corresponds to an etalon produced at the air/pellet interfaces and the position of the etalon depends on the thickness and refractive index of the pellet. The etalon feature is no longer in the collected time frame for pellets thicker than 3 mm, corresponding to the three thickest pellets (3.15, 3.67, and 4.19 mm). Figure 1b shows the corresponding absorption spectra in the 100−20 cm^{-1} spectral range where the reference spectrum is of air. The inset in Figure 1b focuses on the low frequency region from 40 to 20 cm^{-1}.

Inspection of the frequency and time-domain spectra in Figure 1 reveals two prominent features that depend on the thickness of the pellets. First, the frequency-domain spectra from the thinner pellets contain a fringe pattern that corresponds to the etalon created at the pellet/air interfaces. This etalon feature is evident in the time-domain spectrum as noted above. Location of the etalon feature in the time-domain spectrum depends on the physical thickness of the pellet and the refractive index of the material within the pellet. The etalon location creates fringe patterns with different frequencies and amplitudes within the corresponding frequency-domain spectrum. As expected, the frequency of the fringe pattern is lower when the etalon feature is located closer to the main peak in the time-domain spectrum. The amplitude is higher for the thinner pellets as the magnitude of the etalon feature is larger when closer to the main peak. Both of these effects are evident in the spectra shown in Figure 1a and b for pellets of 100, 150, 200, and 250 mg corresponding to thicknesses of 1.07, 1.56, 2.07, and 2.65 mm, respectively.

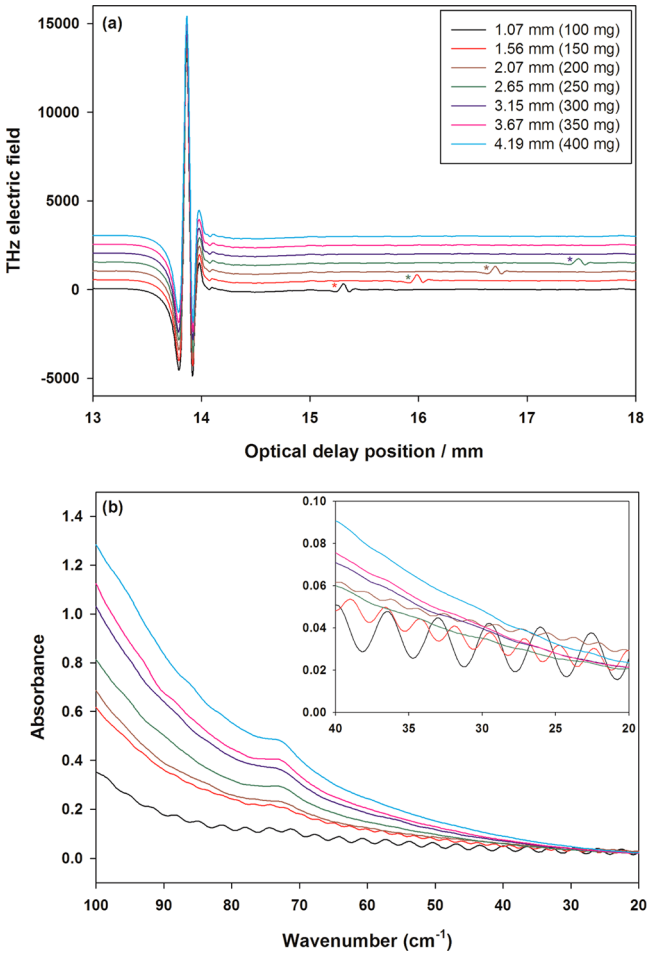

Figure 1. Series of (a) time-domain spectra and the (b) corresponding absorption spectra in the 100−20 cm^{-1} spectral range where the reference spectrum is of air. (Inset) Low frequency region of absorption spectra from 40 to 20 cm^{-1}. For clarity, the time-domain spectra are offset vertically, whereas no offset is included in the absorption spectra. Also, all time-domain spectra are horizontally adjusted to align the maximum point for the main peak signal and an asterisk is used to denote the etalon feature in the time-domain spectrum.

The second prominent feature within these THz spectra corresponds to the nonlinear, sloping spectral baselines, which are evident in all spectra, but are more prominent for the thicker pellets. The magnitude of the baseline increases from low to high THz frequencies for these spectra when air is used as the reference. This sloping baseline is associated with attenuation of THz radiation by scattering within the PE matrix.[19,20] As noted above, air is trapped in the matrix, thereby creating scattering interfaces with mismatched refractive index.

Overall, thinner samples are subject to the etalon fringe pattern throughout the spectrum and thicker samples are subject to loss of radiation through scattering. In general, the Beer−Lambert relationship suggests that thicker samples should provide higher sensitivity and, therefore, lower limits of detection. The impact of increased scattering for thicker samples, however, provides a practical limit on sample thickness as the signal-to-noise ratio degrades with attenuation caused by scattering.

Spectral Features of Lactose. Absorption spectra of lactose were calculated as the ratio of single-beam spectra of the lactose/PE pellets relative to air. These spectra are presented in Figure 2a for a series of pellets prepared with seven different masses of a 10 wt-% preparation of lactose in PE. Consistent with previous reports,[18,31] lactose absorption bands are centered at 90, 75, 63, 45, and 40 cm^{-1} and the magnitude of these absorption peaks increases with more mass (thicker pellets) in the optical path. The etalon fringe pattern and scattering effects described above are superimposed on these absorption features.

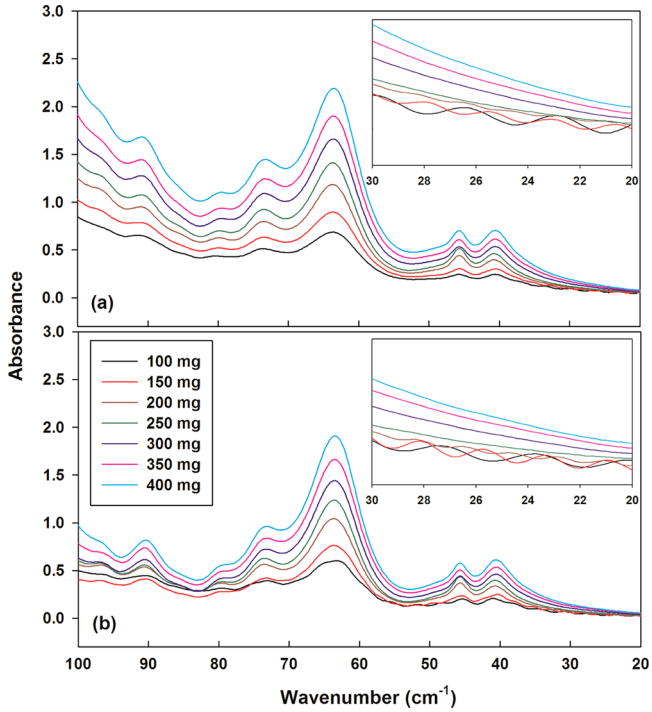

Figure 2. Absorption spectra from 10 wt-% lactose pellets prepared with seven different total pellet masses ratioed to a single-beam spectrum of (a) air and (b) blank PE pellet with the same mass. Each spectrum corresponds to the average of six individual spectra collected by rotating and flipping the pellet, as described in the Experimental Section. (Insets) Spectral features in the 30−20 cm^{-1} spectral range.

A second set of lactose spectra is presented in Figure 2b where the single-beam spectra for the seven different masses of the 10 wt-% lactose/PE mixture are ratioed to a single-beam spectrum of blank PE at the same nominal mass. The same absorption features are evident for lactose when either air or blank-PE is used as the reference medium. Features associated with the etalon and scattering are still present, however, which indicates that blank PE pellets of the same mass do not provide an ideal reference material for these absorption measurements. As detailed above, the thickness of the mixture pellet depends on the respective porosities or compressed densities of lactose and PE. As a result, the physical thickness of the sample and blank PE pellets are different. Optical thickness of these samples depends on a combination of the physical thickness of the pellet as well as the refractive index of the mixture of which the pellet is composed. These effects combine to place the etalon feature in different positions within the two time-domain spectra and result in different frequencies of the etalon fringe pattern in the frequency-domain spectrum.

The nonlinear baseline in the spectra presented in Figure 2b illustrates that scattering is still an issue when blank PE is used

Table 2. PLS Calibration Results for the Determination of Lactose in PE Pellets[a]

	total pellet mass						
	100 mg	150 mg	200 mg	250 mg	300 mg	350 mg	400 mg
	Sample spectra with air reference						
MSEC (wt-%)	0.15	0.10	0.11	0.09	0.04	0.07	0.04
MSEP (wt-%)	0.19	0.15	0.20	0.11	0.06	0.11	0.05
	Sample spectra with blank PE reference						
MSEC (wt-%)	0.15	0.10	0.11	0.09	0.05	0.07	0.05
MSEP (wt-%)	0.20	0.15	0.20	0.11	0.06	0.11	0.06

[a]PLS models created by leave-three-out cross validation. MSEC and MSEP values correspond to the mean standard error of calibration and mean standard error of the cross validation prediction, respectively, obtained from 200 independent repeats of the cross-validation process with a different set of three samples placed in the prediction data set.

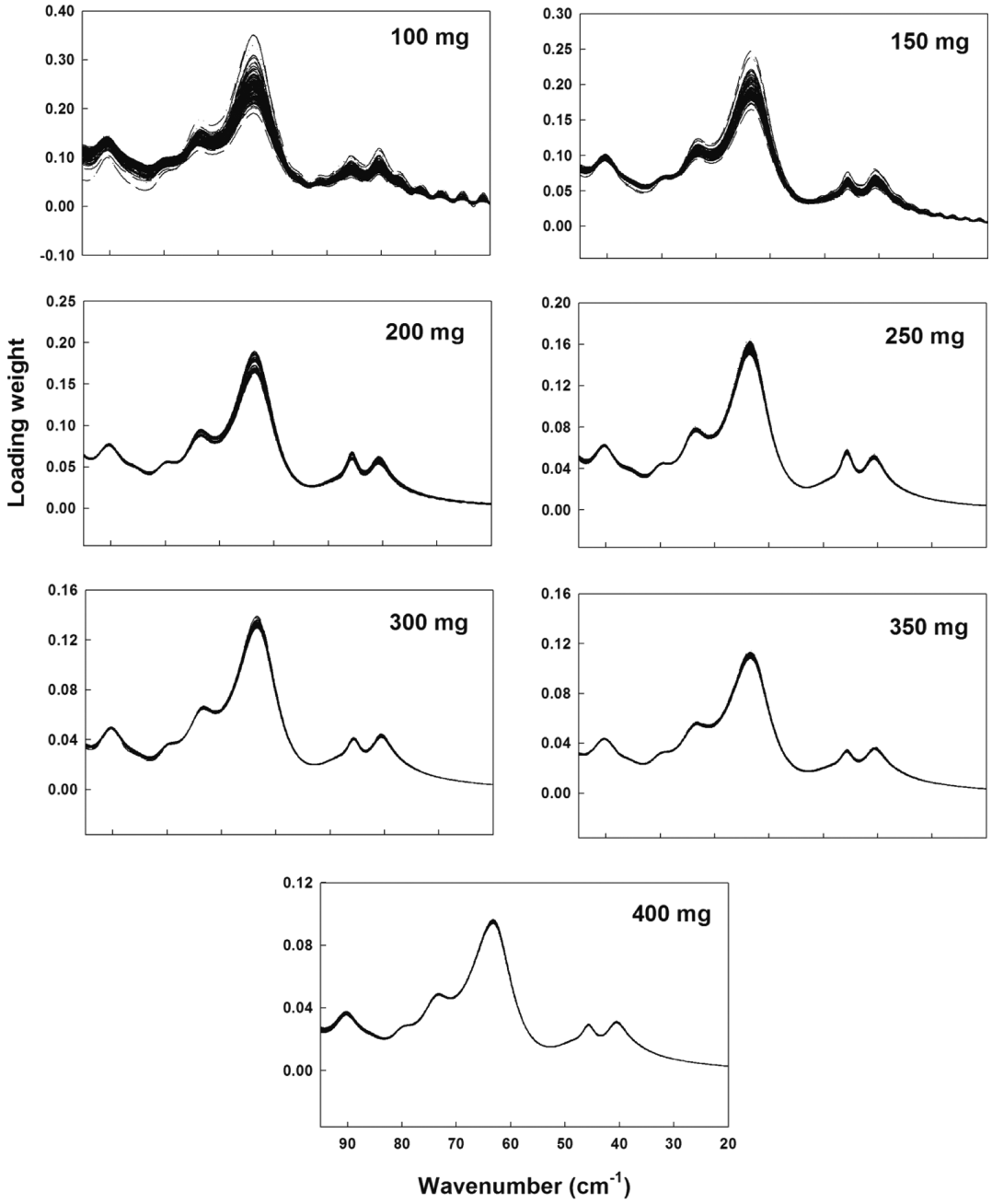

Figure 3. First spectral loading vectors obtained from the 200 leave-three-out cross validation PLS models generated for the measurement of lactose content for the seven different nominal pellet masses.

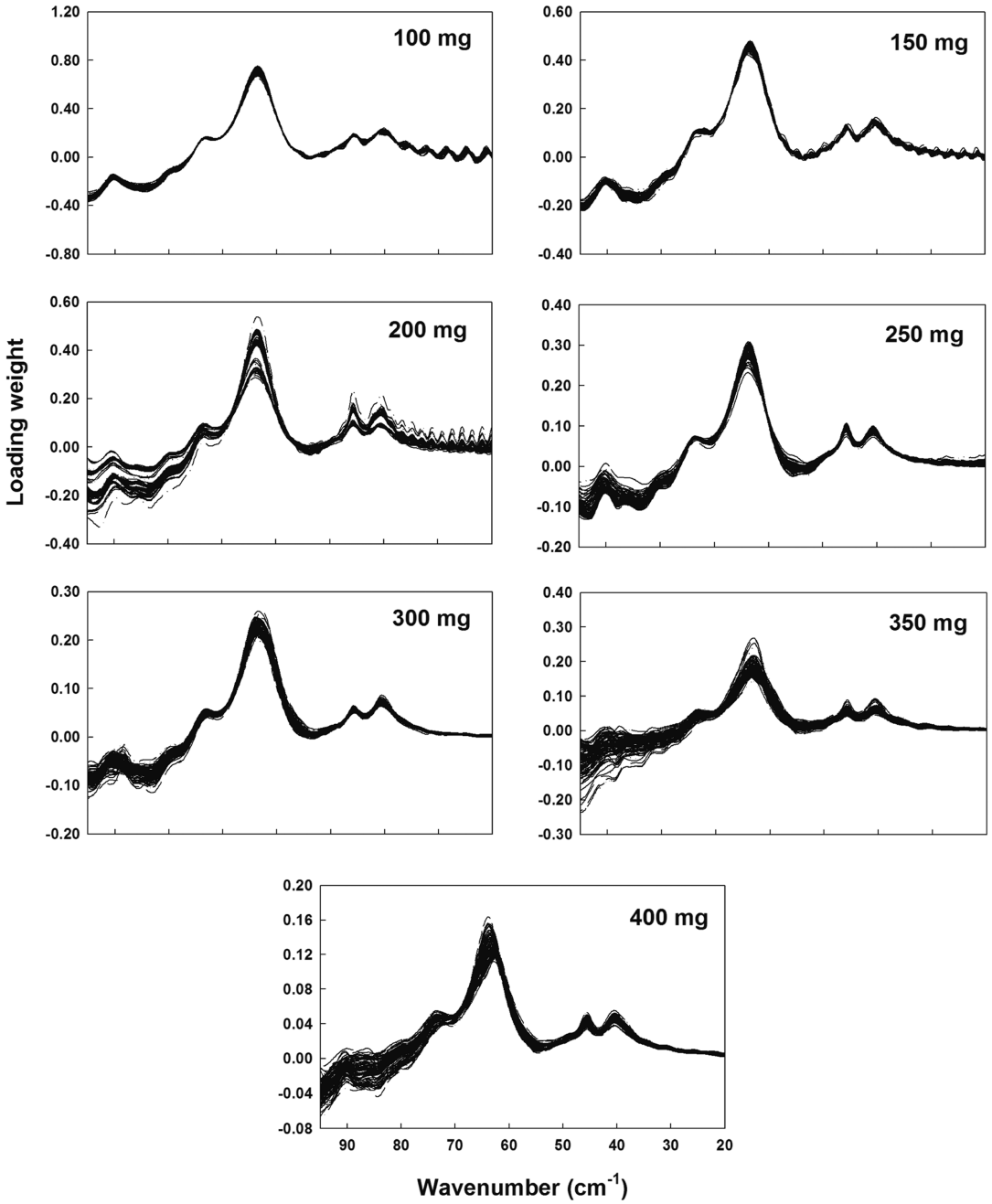

Figure 4. Second spectral loading vectors from the 200 leave-three-out cross validation PLS models generated for the measurement of lactose content for the seven different nominal pellet masses.

as the reference material. Differences in the porosity, particle size, and refractive index between pure PE and lactose-PE mixtures results in differences in the scattering properties of the two pellets (sample vs reference). Still, the magnitude of the scattering effect is noticeably smaller when blank PE is used as the reference material compared to air.

Determination of Lactose Concentration. The confounding nature of the superimposed etalon fringe pattern and the nonlinear baseline effects created by scattering within the pellet suggest that a multivariate calibration method is required to compensate for these sample dependent variations. For this reason, partial least-squares (PLS) regression was used to generate calibration models for the measurement of lactose in the PE matrix. For this experiment, a set of 11 samples was prepared with concentrations ranging from 5 to 15 wt-% with increments of 1 wt-%. A separate PLS calibration model was generated for each pellet mass, resulting in one calibration model for each of seven different sample masses and, correspondingly, seven different nominal pellet thicknesses.

PLS calibration models were created by using a leave-three-out cross validation calculation where all spectra for three samples were placed in the prediction set while all spectra for the eight remaining samples were used to construct the calibration model. The process was repeated 200 times with a different set of three sample spectra used for prediction for each repeat. Each PLS regression model was computed by using a spectral range of 95−20 cm^{-1} and two latent variables (factors). Model performance was judged on the basis of the mean

standard error of calibration (MSEC) and mean standard error of prediction (MSEP) for these 200 cross validation repeats. This process was repeated to establish separate calibration models from spectra with air and blank PE pellets used as the reference spectrum. Overall, 14 separate PLS calibration models were generated and the results are summarized in Table 2. Statistical testing indicates no significant difference at the 95% confidence level when using air or blank PE as the reference material.

The values in Table 2 indicate that model performance depends on the mass or thickness of the pellets. Larger errors are observed for the thinner pellets and the trend in model error decreases before a minimum is reached at a thickness of 3.15 mm, corresponding to a sample mass of 300 mg. Errors increase for the 350 mg sample and then fall again for the thickest pellets composed of 400 mg of the sample.

Differences in the performance of these models were investigated by examining the spectral loadings (eigenvectors) for each PLS calibration model. Spectral loadings are presented in Figures 3 and 4 across all the models for the first and second latent variables, respectively. In both cases, spectral loadings are superimposed for each of the 200 cross validation PLS calibration models determined for each pellet mass. These loading vectors correspond to models generated from spectra with air as the reference material.

For the first latent variable (Figure 3), variation across models is most apparent for the thinner pellets and the degree of variation decreases as the mass of the pellet increases. The large variation of spectral loading vectors for the thinner samples is a direct indication that the corresponding spectra are less reproducible. This observation is consistent with the larger etalon fringe patterns noted above for the thinner samples. Variations for the second latent variable (Figure 4) indicate the opposite effect where the magnitude of the variance is smallest for the thinner samples and increases with sample thickness. As expected, the overall magnitude of the variance is lower for the second factor, but the degree of variance is not uniform for the different pellet masses. The larger variations noted for the thicker samples and at the higher frequencies (100−70 cm^{-1}) are consistent with the impact of scattering effects, as illustrated in Figures 1 and 2.

Accuracy of lactose measurements are evident by the concentration correlation plots provided in Figure S3 (Supporting Information) for each PLS calibration model. In all cases, predicted concentrations correlate well with the prepared concentration with uniform scatter about the ideal unity line. Linear regression analysis for the correlation between predicted versus prepared lactose concentrations indicates functioning models for each pellet mass. These results are summarized in Table S2 (Supporting Information). Overall, similar analytical performance is noted for each of the calibration models across different masses of samples. As indicated by the MSEC and MSEP values reported in Table 2, lower measurement errors are evident for pellets prepared with 300 and 400 mg of sample. As illustrated in Figures 3 and 4, the degree of spectral variance is lowest for spectra collected with the 300 mg samples. The concentration correlation plot for the best model when using 300 mg samples is characterized by a slope of 1.006 ± 0.004, a y-intercept of −0.038 ± 0.043 wt-% and a coefficient of determination (R^2) of 0.999.

CONCLUSION

Sound calibration models are demonstrated for quantitative THz absorption measurements for a simple sample matrix composed of solid lactose particles suspended within a pellet of compressed solid PE. Physical thickness of these pellets represents a critical experimental parameter that impacts model performance.

Thinner pellets are sensitive to etalon effects created as the transmitted THz radiation propagates through the air/sample and sample/air interfaces of the pellet. Spectral artifacts induced by the etalon effect cannot be easily removed by ratioing to a reference spectrum. Elimination of the etalon effect demands the use of a reference spectrum collected from a blank pellet with the exact same optical thickness, as obtained from a combination of the physical thickness and refractive index of the sample. Results presented here illustrate how the physical thickness of the pellet is sensitive to its composition. Differences in the porosity of the compressed materials, PE and lactose in this case, result in different physical thicknesses for different relative amounts of lactose. The final pellet thickness can be predicted as the linear combination of the individual components. Still, finding a suitable blank reference pellet to cancel the etalon effect requires previous knowledge of the composition of the pellet or a sensitive method for measuring pellet thickness.

THz spectra collected from thicker pellets are sensitive to scattering effects generated from refractive index mismatches at the air/lactose and air/PE interfaces within the pellet structure. Again, the magnitude of the scattering effect depends on the thickness of the pellet, which depends on its composition and is not simple to eliminate. Improvements in quantitation can be realized by reducing the scattering properties of the sample. Two possible strategies include decreasing the amount of air trapped within the pellet matrix, possibly by heating the PE preparation prior to pellet formation, and reducing the particle size of the PE and analyte preparations.[20]

Overall, consideration of the thickness of solid samples can improve analytical performance. Although PLS calibration models can be used to reduce the impact of these sources of variation, examination of the impact of sample thickness on spectral variance can provide a means to optimize this parameter for individual sample matrixes.

ASSOCIATED CONTENT

Supporting Information

Supplementary tables and figures. This material is available free of charge via the Internet at http://pubs.acs.org.

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Notes

The authors declare no competing financial interest.

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